

Composing Context Compression for Long-Horizon LLM Agents

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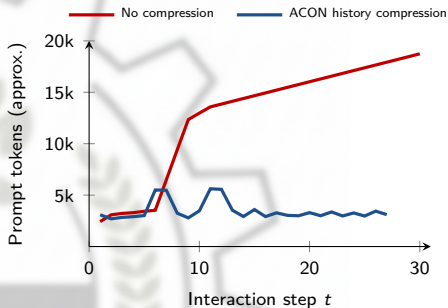
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Context Grows Without Bound



Our own runs, AppWorld task 34d9492_2 (both arms, 30 iterations). Baseline read from the agent's conversation log; the compressed curve reconstructed from the 17 compressor-call records of the same task. Tokens estimated at 4 characters/token. Peak: **18.7k** vs. **5.6k**.

Two bottlenecks

- **Memory** — the KV cache scales with sequence length
- **Distraction** — long, partly stale context degrades decisions

The tension

Compression must be *aggressive* (to save tokens) and *precise* (one lost file path or API token derails an entire trajectory).



Methods We Build On, and the Field

ACON — prompt-level, generative

- An LLM *rewrites* history when $|h_t| > T_{\text{hist}}$, shortens observations when $|o_t| > T_{\text{obs}}$
- Guideline optimised *in natural language* by contrasting uncompressed successes with compressed failures
- Agent frozen; compressor distilled into a small model
- AppWorld: 56.5% vs. 56.0% acc., peak tokens $\downarrow$ 26–54%

SAC — KV-level, soft

- Selects *anchor tokens*, marks them with a learned embedding
- Bidirectional attention lets anchor KV absorb the whole context
- No autoencoding pretraining required
- MRQA, Llama-3.2-1B at 15 $\times$: 54.95 ID F1 (EPL 51.52)

Property	ACON	SAC	SWE-Pruner	LLMLingua	MEM1/MemAgent	Proposed
Agent weights frozen	Yes	Yes	Yes	Yes	No	Yes
History compression	Yes	No	No	Yes	Merged	Yes
Observation compression	Yes	No	Yes	Yes	Merged	Yes
Composition studied	Partly	No	No	No	No	Yes
Intrinsic quality metric	No	No	No	No	No	Yes (CFR)

Every existing method compresses *one* kind of context — how the stages **interact** is unstudied.



Result 1: ACON on AppWorld

Setup: one RTX 2000 Ada (16 GB), Ollama; agent qwen2.5:14b, compressor qwen2.5:7b;

Run	Tasks	Max iter.	Solved	Avg iter.	Agent input tok.	Wall clock (s)
Baseline, no compression	8	12	1	11.00	269,935	953.9
History compression ($T_{\text{hist}}=512$)	8	12	2	10.75	232,749	882.8
Observation compression ($T_{\text{obs}}=256$)	8	12	1	11.00	261,422	873.6
Baseline, full split	38	30	12 (31.6%)	23.7	2,972,376	7312
Baseline, matched subset	18	30	7	22.22	1,283,906	—
History compression, matched subset	18	30	9	19.50	969,389	—

- 18 matched tasks: **9 solved vs. 7, 24.5% fewer** agent input tokens
- Not uniform: solved 5 the baseline failed, failed 3 it solved
- No latency overhead at this scale — compressed runs finished sooner

Caveats carried into every claim

Thresholds 512/256 (paper: 4096/1024, they rarely fire here); 4096-token serving window truncates long baseline prompts; one seed per setting.



Result 2: SAC Reproduction and Pilot

SAC reproduces to within 0.2 F1

15×: **54.98** ID / **39.25** OOD (published 54.95 / 39.26) 5×: **63.70** / **47.71** (published 63.63 / 47.72)

Run	ID F1	OOD F1	Δ ID vs. R0 (95% CI)	Δ OOD vs. R0 (95% CI)
R0: SAC fine-tuning (matched control)	51.37	38.62	control	control
R1: + uniform KL (ablation)	51.06	38.17	−0.32 [−1.02, +0.44]	−0.45 [−2.11, +1.44]
R2: + FGD (gated KL, $\lambda=1$)	51.15	38.47	−0.23 [−0.95, +0.47]	−0.14 [−2.04, +1.77]
R3: + TGA + FGD	60.03	44.71	+ 8.66 [+7.66, +9.60]	+ 6.10 [+3.62, +8.96]
<i>Published SAC 15× (20,000 steps)</i>	<i>54.95</i>	<i>39.26</i>		

- **FGD is a null result.** Its intervals contain zero. A diagnostic explains why: the zero-shot full-context teacher had *higher* gold-answer NLL than the compressed student on **90.5%** of 2,000 training examples — there is little to distil.
- **TGA + FGD wins**, beating the published 20,000-step checkpoint after a quarter of the budget — but the question is seen during compression, so the memory is *not reusable* across questions. This is a separate query-aware setting, not a drop-in improvement.



Three Gaps in Composition and Evaluation

(a) Tokens $\neq$ time

The compressor is itself an autoregressive LLM call, and rewriting history **invalidates the KV cache**, forcing the prefix to be re-encoded.

ACON's own medians (Qwen3-14B, one A100):

No compression	73.24 s
History	87.68 s
Observation	101.92 s

A token objective $C(\pi)$ misses this entirely.

(b) Schedule confound

The trigger set depends on the pipeline itself:

$$\mathcal{K}(\pi) = \left\{ t : |h'_{t-1} \oplus (o'_t, a_t)| > T_{\text{hist}} \right\}$$

Since $|o'_t| \leq |o_t|$, compressing observations first shifts *when* history compression fires: $\mathcal{K}(\pi_{ho}) \neq \mathcal{K}(\pi_h)$.

$$I = \Delta(\pi_{ho}) - \Delta(\pi_h) - \Delta(\pi_o) = -9.5$$

I mixes composition with the shifted schedule, so it is *not* evidence that composition harms.

(c) Quality measured indirectly

ACON and SAC judge a compressor **solely** by downstream success:

$$h'_t \rightarrow \text{rollout} \rightarrow R$$

A failure could mean the compressor dropped a needed fact, *or* that the agent reasoned badly. Neither method measures whether the facts the agent *will need* survive, so failures **cannot be attributed**.



Problem Statement

Informally

Given a long-horizon agent with **frozen weights**, a history compressor and an observation compressor, determine how the two stages should be **composed and scheduled** so that the token savings of each are retained without lowering task success — holding the compression schedule fixed, and measuring compression quality *directly*.

Agent (POMDP, frozen θ):

$$a_t = M(h'_{t-1}, o'_t; \theta)$$

Compression operators:

$$o'_t = \begin{cases} f_o(o_t, h'_{t-1}) & |o_t| > T_{\text{obs}} \\ o_t & \text{else} \end{cases}$$

$$h'_t = \begin{cases} f_h(x_t) & t \in \mathcal{K} \\ h'_{t-1} \oplus (o'_t, a_t) & \text{else} \end{cases}$$

Two composition orders (ACON uses x^{seq}):

$$x_t^{\text{seq}} = h'_s \oplus (o'_{s+1}, a_{s+1}, \dots, o'_t, a_t)$$

$$x_t^{\text{dec}} = h'_s \oplus (o_{s+1}, a_{s+1}, \dots, o_t, a_t)$$

Cost, peak context and latency:

$$C(\pi) = \sum_{t=1}^T (|h'_{t-1}| + |o'_t|)$$

$$L(\pi) = \sum_t \ell_M + \sum_{t \in \mathcal{K}} \ell_{f_h} + \sum_{|o_t| > T_{\text{obs}}} \ell_{f_o}$$

Formal objective:

$$\pi^* = \arg \max_{x, \mathcal{K}} \mathbb{E}[R(\pi)] - \lambda \mathbb{E}[C(\pi)] - \mu \mathbb{E}[L(\pi)]$$

subject to

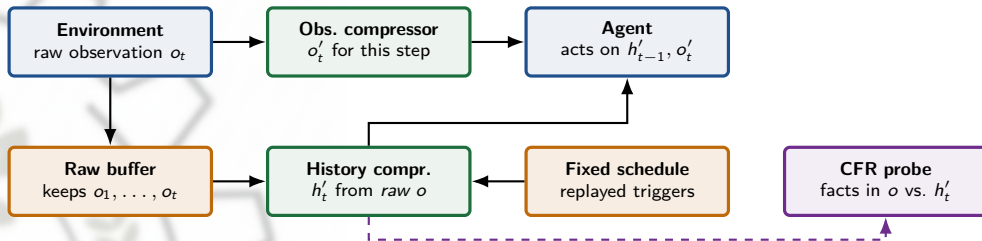
$$\mathbb{E}[R(\pi)] \geq \mathbb{E}[R(\pi_{\emptyset})] - \varepsilon$$

$$C(\pi_{ho}) \leq \min\{C(\pi_h), C(\pi_o)\}$$

$$\mathcal{K}(\pi_{ho}) = \mathcal{K}(\pi_h) \text{ when comparing them}$$



Proposed Pipeline



1. Decoupled composition order

The agent still gets the short o'_t , but the **raw** o_t is kept in a side buffer and f_h receives x_t^{dec} . If f_o only deletes content then $V(o'_t) \subseteq V(o_t)$, so facts dropped by the observation stage *can still reach* the history summary.

2. Controlled compression schedule

Record $\mathcal{K}^h = \mathcal{K}(\pi_h)$ from the history-only run and **replay** it: $\mathcal{K}(\pi_{ho}) := \mathcal{K}^h$. With the schedule fixed, π_h and π_{ho} differ *only* in f_o , so I now measures composition alone.



Critical-Fact Retention (CFR)

Definition

Let $V(\cdot)$ extract state variables — access tokens, entity IDs, file paths, numeric values — and let $a_{1:T}^{\emptyset}$ be the actions of the **uncompressed reference run**. The facts at stake in trigger t , and the retention rate, are

$$F_t = V(x_t) \cap \bigcup_{t'} V(a_{t'}^{\emptyset}), \quad \text{CFR}(\pi) = \frac{\sum_{t \in \mathcal{K}} \sum_{v \in F_t} \mathbf{1}[v \sqsubseteq h'_t]}{\sum_{t \in \mathcal{K}} |F_t|},$$

where $v \sqsubseteq h'_t$ means an *exact, verbatim* occurrence of v in h'_t .

Why it is cheap

- $V(\cdot)$ is a set of pattern extractors for AppWorld's tokens, IDs, paths and numbers
- Computed **offline** for every compressor call
- **No additional agent rollouts** required

What it lets us test

- Rank correlation $\rho(\text{CFR}, R)$ — does fact retention actually predict task success?
- CFR under x^{seq} vs. x^{dec} — does decoupling really retain more?
- Attribute a failure to **information loss** rather than to the agent



Complete the reproduction

- Finish the 38-task history-compression run
- Run observation and combined history + observation compression
- Add the paper thresholds (4096/1024) and the llama3.1:8b compressor on the same split

Implement the proposed pipeline

- Raw-observation buffer (x^{dec})
- Schedule replay $\mathcal{K}(\pi_{ho}) := \mathcal{K}^h$
- CFR extractor; then estimate I under *both* orders with the schedule held fixed

Measure latency directly

Log $\ell_M, \ell_{f_h}, \ell_{f_o}$ per call to decompose $L(\pi)$, including **prefix re-encoding** after each history compression.

Strengthen the evidence

- At least **three seeds** per setting
- Raise the served context window above 4096 tokens so baseline prompts are not truncated
- For SAC $\times$ ACON: run **TGA alone**, plus the $5\times$ and $51\times$ ratios



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Thank You

Questions?

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